

Rover Reliability Trial

Compete for the best combination of mission success, time, accuracy, and fault recovery.

Advanced	2-3 class periods	Competition
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Engineering Context

A rover design review requires objective evidence of reliability. Each team will complete repeated missions under the same conditions and receive one controlled fault scenario.

What You Will Practice

- Define reliability metrics
- Run repeated controlled trials
- Calculate success rate
- Respond to a fault without unsafe behavior

What You Need

- Rover and mission course
- Standard payload
- Timer
- Fault scenario cards
- Data sheet
- Engineering notebook

Student Instructions

- 1 Review the scoring categories and identify how each will be measured.
- 2 Complete a preflight inspection.
- 3 Run three baseline missions and record time, path error, and task success.
- 4 Receive one instructor-approved fault such as blocked sensor view, shifted target, or low-traction surface.
- 5 Run two fault-condition missions.
- 6 Calculate overall success rate and average completion time.
- 7 Explain how the system detected or failed to detect the fault.
- 8 Make one approved change and complete a final demonstration.

Engineering Requirements

- Same code for official baseline trials
- Complete required mission task and return safely
- Record every trial, including failures
- Fault condition must not create unsafe motion
- Score includes reliability, not only fastest time

Constraints & Safety

- Emergency stop required
- Instructor controls fault setup
- Do not alter another team’s equipment
- Keep all observers outside the course

What You Submit

- Preflight checklist
- Five-trial data set
- Reliability calculations
- Fault analysis
- Final demonstration result

Reflection Questions

- 1 What was the most frequent failure mode?
- 2 Did the fastest trial represent normal performance?
- 3 How should reliability be reported with only a few trials?
- 4 What design change would reduce operator dependence?

Engineering Tip Use identical start configuration for every official trial to make the comparison fair.	Common Mistake Deleting failed trials creates a misleading reliability result.
Stretch Goal Complete ten consecutive missions and calculate a larger-sample success rate.	Career Connection Flight-test and reliability engineers analyze repeated trials, anomalies, and fault responses before operational approval.